SUPPORTING ADULT GUIDANCE

Showing solutions on a number line

Inequalities · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson asks learners to **draw and read** solution sets on a number line themselves, after two lessons in which the lesson drew them. Learners move between four representations (words, symbols, number line, list of whole numbers), solve double-ended inequalities by operating on all three parts, find where two inequalities overlap, and build pictures with a tool that reads their picture back to them.

Driving Question: *How can one picture show every answer at once?*

The mathematics, in plain terms

On a number line, a closed (filled) circle marks an included endpoint (≤, ≥) and an open circle an excluded one (<, >). A ray shows an unbounded set; a segment shows a bounded one, written as a double-ended inequality such as $-2 < x \leq 3$. A double-ended inequality is solved by applying each inverse operation to all three parts. The set satisfying two inequalities at once is the intersection (overlap) of their solution sets, which may be empty.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Drawing a one-sided inequality

Draw $x \geq -1$.

1. The boundary is –1. Is –1 included? ≥ means yes.	filled circle at –1
2. x is greater than –1, so the set goes to the right.	arrow to the right
3. Whole numbers in the set:	–1, 0, 1, 2, …

A filled circle at –1 with an arrow to the right.

Example 2 — Solving and drawing a double-ended inequality

Solve $-1 < 2x + 3 \leq 9$ and draw the answer.

1. Subtract 3 from all three parts.	$-4 < 2x \leq 6$
2. Divide all three parts by 2.	$-2 < x \leq 3$
3. Open circle at –2 (not included), filled at 3 (included).	whole numbers –1, 0, 1, 2, 3

$-2 < x \leq 3$: a segment from an open circle at –2 to a filled circle at 3.

Let the read-back do the correcting

In the picture maker, the lesson reads back what the learner's picture actually says, in symbols, words and whole numbers. Encourage learners to compare that read-back with what they intended before pressing Check. If you are asked for help, point to the read-back rather than to the answer: 'What does your picture say? Is that what you meant?'

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Four ways to say it: find the card that does not match the other three.	~7 min
Movement	The three-part balance: solve double-ended inequalities by doing the same to every part.	~10 min
Reflection	Where two rules overlap: tap the whole numbers that obey both, then name the overlap.	~10 min
Creativity	The picture maker: draw solution sets and have them read back, including real situations.	~13 min
Rest	Picture an arrow running on for ever, and the circle where it starts.	~2 min

Nutrition	Intellectual: representation — saying one idea several ways and choosing the clearest.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Open and filled circles the wrong way round.
Ask: 'Is the boundary number itself allowed? Test it in the inequality.' For $x < 4$: is $4 < 4$ true?

Arrow pointing the wrong way, especially when the inequality is written backwards ($5 > x$).
Ask the learner to read it starting with x: 'x is less than 5', then test a number on each side.

Listing whole numbers that include an excluded boundary, or missing an included one.
Check the two ends of the list against the circles, one at a time.

Doing a step to the middle part only in a double-ended inequality.
'What you do to the middle, you do to both ends. Which part did you forget?'

Assuming two rules always overlap.
Draw both pictures one above the other. 'Is there any number under both pictures?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: placing whole numbers, including negatives, on a number line, and solving inequalities from Lesson 2. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner draws $x \geq 2$ with an open circle at 2. The aim is a question that lets them find the slip themselves.

- "What does your picture say? Read it back to me in words."

- “Is 2 itself allowed by $x \geq 2$? Try it: is $2 \geq 2$ true?”
- “What does an open circle mean? And a filled one?”
- “Which whole numbers does your picture include? Does that match the inequality?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why practise reading as well as drawing?

Exam questions and real-life information present the picture and ask for the inequality as often as the other way round. Reading builds the same understanding from the other side.

Why does the three-part balance refuse negative divisors?

Dividing every part by a negative reverses both symbols and swaps the ends round, which is a later step. Every inequality in this lesson can be solved with positive numbers.

Can there be more than one correct picture?

For whole numbers, yes: $-1 \leq x \leq 2$ and $-2 < x < 3$ contain the same whole numbers. The lesson draws attention to this when it happens.

What comes next?

Lesson 4 moves inequalities onto a graph, where $x > 2$ becomes a whole region, and overlapping regions play the part that overlapping number lines played today.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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